Module : 4

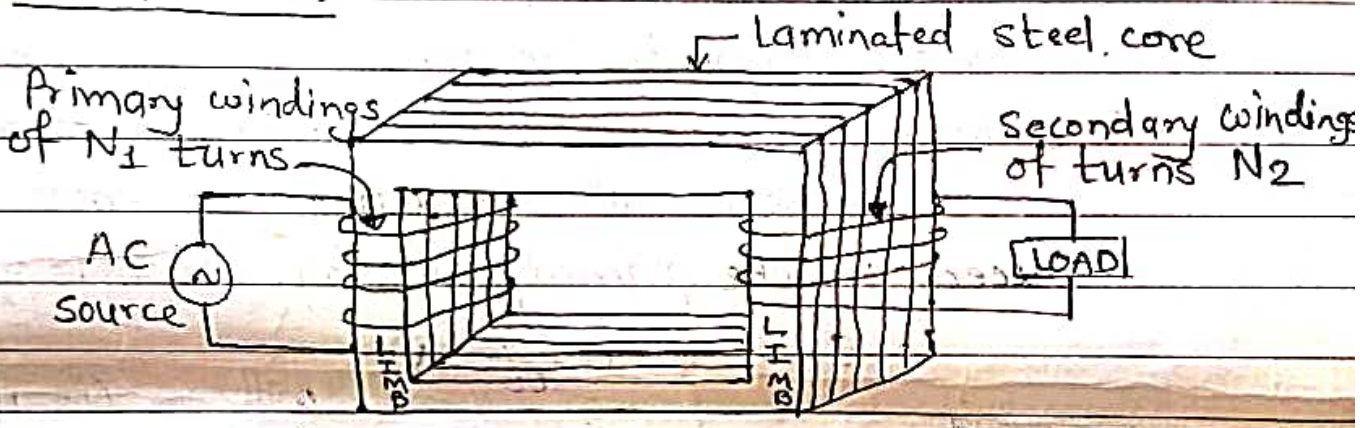
Single-phase Transformer

Introduction:-

Transformer is a static device which can transfer electrical energy (AC) from one circuit to another circuit without any change of frequency. It can increase or decrease the voltage but with a corresponding decrease or increase in current.

It works on the principle of mutual induction.

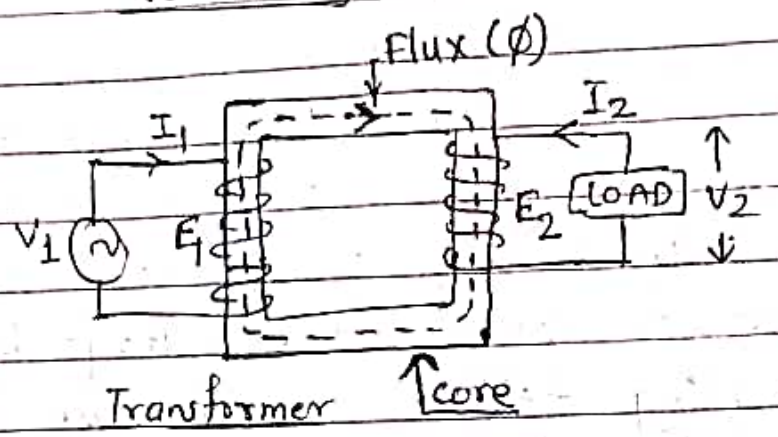
Construction :-

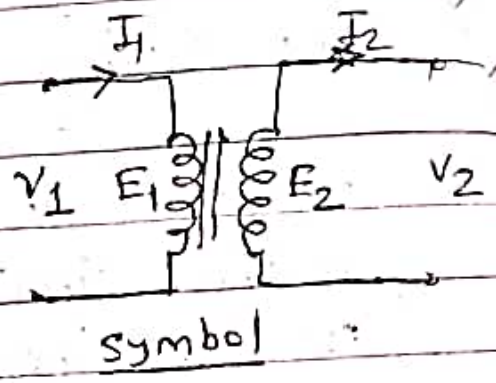


A transformer mainly consists of an iron core, and two inductive coils or windings placed on a common core. One winding is connected to the source of the electrical energy (called primary winding), while the other is connected to the load (called secondary winding). These windings are of different number of turns, wound around the two opposite limbs of the core. Each winding is properly insulated from each other as well as core.

A core is rectangular in shape and is laminated. The core is constructed of stacks of laminated silicon steel to provide a continuous magnetic path to carry alternating flux.

Working Principle :-





When an alternating voltage V_1 is applied to primary coil or winding of a transformer, an alternating current I_1 flows through it producing alternating flux in the core. This flux links with the primary winding and according to Faraday's law of electromagnetic induction, an emf e_1 is induced in the primary winding, which is given by,

$$e_1 = -N_1 \frac{d\phi}{dt} \quad V$$

where N_1 is the number of turns in the primary winding. The emf induced in primary windings is called self induced emf.

Assuming leakage flux to be negligible, almost whole flux produced by the primary winding links with the secondary winding. Hence, an emf e_2 is induced in it, which is given by

$$e_2 = -N_2 \frac{d\phi}{dt}$$

where N_2 is the number of turns in secondary winding. The emf induced in secondary windings is called mutually induced emf. If the secondary winding is closed through the load, a current I_2 flows through the secondary coil. Thus, energy is transferred from the primary coil to secondary coil.

EMF Equation of a transformer

Let the equation of the sinusoidal alternating flux produced in the core be

$$\phi = \phi_m \sin \omega t$$

According to Faraday's law of electromagnetic induction, the self induced emf in primary winding is

$$e_1 = -N_1 \frac{d\phi}{dt}$$

$$\therefore e_1 = -N_1 \frac{d}{dt}(\phi_m \sin \omega t)$$

$$\therefore e_1 = -N_1 \phi_m \omega \cos \omega t$$

$$\therefore e_1 = N_1 \phi_m \omega \sin(\omega t - 90^\circ) \quad \text{—— (I)}$$

[Note: e_1 lags ϕ by 90°]

Comparing equation (I) with standard sinusoidal form, i.e. $e = E_m \sin(\omega t \pm \phi)$, Maximum value of the induced emf is given by,

$$E_m = N_1 \phi_m \omega$$

Hence, rms value of the induced emf in primary winding is given by.

$$E_1 = \frac{E_m}{\sqrt{2}} = \frac{N_1 \phi_m \omega}{\sqrt{2}} = \frac{N_1 \phi_m 2\pi f}{\sqrt{2}}$$

so, $\boxed{E_1 = 4.44 f N_1 \phi_m}$ V —— (II)

where ϕ_m is the maximum flux in Wb and f is the supply frequency.

Similary, rms value of the induced emf in secondary winding is, $\boxed{E_2 = 4.44 f N_2 \phi_m}$ V —— (III)

Equations II & III are called emf equations

$\boxed{\phi_m = B_m A}$ where, A is cross-section area of core (m^2)

B_m is maxi. flux density in core (in Tesla, T)

Transformation Ratio (K)

Transformation ratio is denoted by K

$$K = \frac{E_2}{E_1} = \frac{N_2}{N_1} = \frac{V_2}{V_1} = \frac{I_1}{I_2}$$

For step up transformer, $(E_2 > E_1)$ $N_2 > N_1$ so $K > 1$

for step down transformer, $(E_2 < E_1)$ $N_2 < N_1$ so $K < 1$

Practical (Actual) and Ideal transformer

Practical transformer	Ideal transformer
1. There are copper & core losses	1. There is no loss.
2. There is leakage of flux	2. There is no leakage of flux
3. It's windings contains resistance	3. It's windings consist of pure inductive coil.
4. Voltage regulation is never 0%	4. Voltage regulation is 0%
5. Efficiency is 93 - 97 %	5. Efficiency is 100%
6. All constructed transformers are practical transformer	6. It is impossible to construct an ideal transformer.

Transformer losses :-

The main losses in transformer are copper loss and iron (core) loss.

1) Copper loss (W_{cu}) -

The loss that takes place due to winding resistance is called copper loss (as windings are made of copper). If R_1 and R_2 are resistances of primary and secondary winding respectively, then

Total copper loss, $W_{cu} = I_1^2 R_1 + I_2^2 R_2$ [W]

Copper loss is a variable loss as varies with square of the primary and secondary currents which depends upon the load on the transformer. Copper loss at half load is one-forth of that at full load. At no load, copper loss is negligible. Copper loss can be calculated by conducting short-circuit test on transformer.

2) Iron loss or core loss (W_i) -

The hysteresis loss and eddy current loss take place inside the magnetic core of transformer. These two losses together called core loss. Since the core is made of iron, it is also termed as iron loss. The core loss is proportional to the applied voltage. Normally, the applied voltage is constant, so the core loss remains constant for all loads. Core loss can be calculated by conducting open-circuit test (OC test) on transformer.

a) Hysteresis loss - Since the supply to the primary winding is alternating, the core gets alternately magnetized and demagnetized causing loss of energy. This loss is given by,

$$W_h = \eta B_m^{1.6} f V \quad (W)$$

Where, W_h = hysteresis loss in W

B_m = maximum value of flux density in tesla (T)

f = frequency in Hz

η = steinmetz constant

V = volume of the material in m^3

The silicon steel material used for manufacturing the core where hysteresis loss is very low.

b) Eddy current loss – This loss is due to eddy current induced in the transformer core and is seen in the form of heat in the core, This loss is given by,

$$W_e = K B_m^2 f^2 t^2 \quad (W)$$

Where t is thickness of the lamination. To minimize this loss, the core is not manufactured as solid core but as stack of laminations.

KVA rating –

The output of a transformer is expressed in KVA (i.e. kilo-volt ampere). The rated transformer output is limited by heating and hence losses in the transformer, i.e. copper and core loss. These losses depend upon the voltage and current, and are almost unaffected by the power factor of the load. Therefore, the transformer rated output is expressed in KVA and not in KW.

The KVA rating is given by,

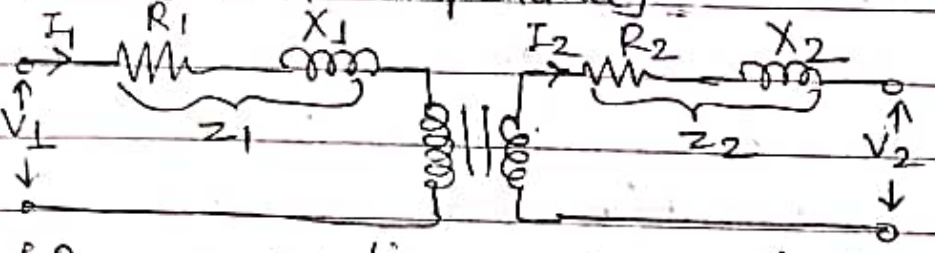
$$\text{KVA rating} = \frac{E_1 (I_1)_{FL}}{1000} = \frac{E_2 (I_2)_{FL}}{1000}$$

where, $(I_1)_{FL}$ = full-load primary current
$(I_2)_{FL}$ = full-load secondary current

$$(I_1)_{FL} = \frac{KVA \times 1000}{E_1}$$

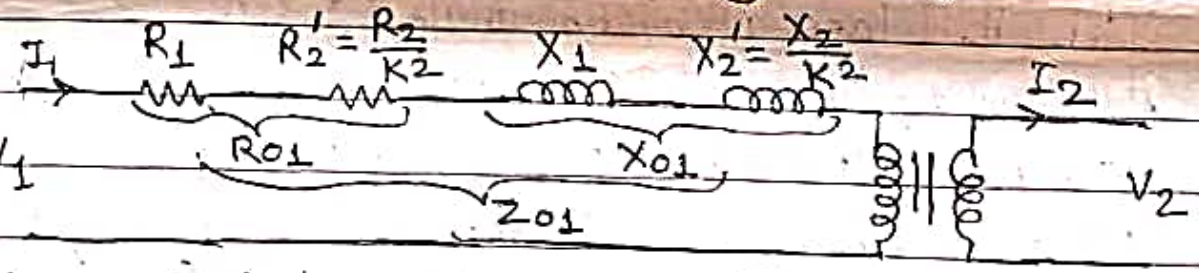
$$(I_2)_{FL} = \frac{KVA \text{ rating} \times 1000}{E_2}$$

* Transformer Parameters [winding resistance, leakage reactance and Impedance]



R_1 & R_2 are winding resistances of primary & secondary resp.
X_1 & X_2 are leakage reactances of 〃 〃
Z_1 & Z_2 are impedances of 〃 〃

→ Shifting of parameters to primary side



The equivalent resistance as referred to primary,

$$R_{01} = R_1 + R_2' = R_1 + \frac{R_2}{K^2} \quad (\Omega)$$

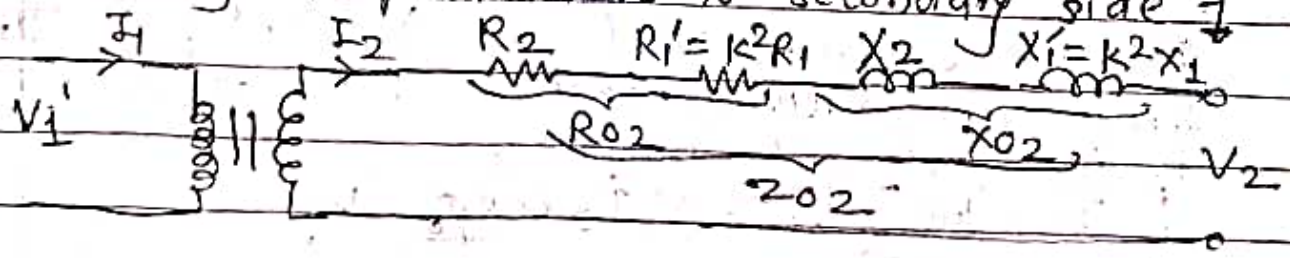
The equivalent reactance as referred to ~~secondary~~ primary

$$X_{01} = X_1 + X_2' = X_1 + \frac{X_2}{K^2} \quad (\Omega)$$

The equivalent Impedance as referred to primary

$$Z_{01} = \sqrt{R_{01}^2 + X_{01}^2} \quad (\Omega)$$

→ shifting of parameters to secondary side



Equivalent resistance referred to secondary

$R_{02} = R_2 + R_1' = R_2 + K^2 R_1$ (Ω)

Equivalent reactance referred to secondary

$X_{02} = X_2 + X_1' = X_2 + K^2 X_1$ (Ω)

Equivalent Impedance referred to secondary

$Z_{02} = \sqrt{R_{02}^2 + X_{02}^2}$ (Ω)

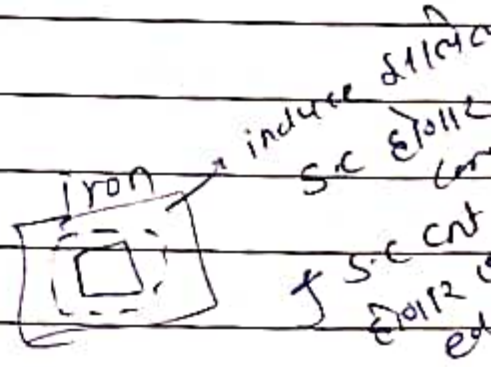
* Losses in X'mer

1) Core losses / iron loss — Constant
2) Copper losses

1) Core losses / iron losses

> Core losses are produced in a X'mer core

> two types iron loss
① eddy losses
② Hysteresis losses



① eddy losses

> The flux is alternating ~~that~~ it link with the X'mer iron core

reduce eddy crt loss

> eddy crt loss are minimised by using lamination of metal to make a metal core > the lamination are seperated by an insulating material > the plane of laminan must be arranged parale to the magnetic field so that they cut the eddy crt path

> when the flux link with the iron core according to the faraday law of electromagnetic induction ther induce emf in iron core

> & these induced emf in iron core there is short circulating crt in the iron core that short ckt crt is called edy crt

> losses is producing because of These eddy crt to called eddy crt losses

② Hysterisis loss

How to reduce: making the core from soft iron or iron alloy to allow easy magnetization & demagnetisation and reduce Hysteresis loss

> As we know that the nature of flux is alternating so before n

befor magnetizing of iron core all the molicular magnet & dipole are random direction

> when core get magnetize all the molicules magnet & dipole are alining the uniform direction.

> as magnetic field is reversible or alternating so dipole has change it's oriantation after every half cycle

> so changing flux it's means changing dir direction of molicular magnet direcn dipole

> so that changing it's orientation after every half cycle some energy loss are take place

Then thse called hysteresis loss.

(B) Coopper loss

> So Copper loss are take place in secondary winding & pry wdg.

> cnt flowing in pry as well as sec wdg pry wdg & sec wdg having resistances

Pry wdg resistance — R_1

Sec wdg resistance — R_2

> cnt flows through a resistor again power loss take place

$I_1^2 R_1$ — pry loss

$I_2^2 R_2$ — sec loss

addn of two loss these Cu loss

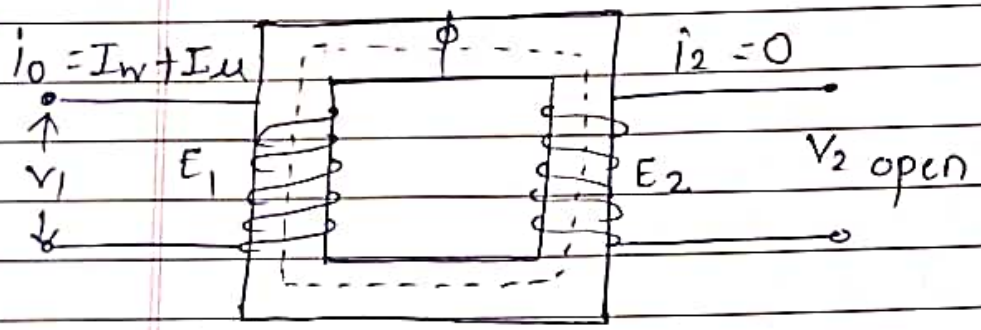
Copper losses reduces

Transformer with high vtg & cnt rating requires condr of large crossection to help minimizing cu losses

I_w : Iron loss component

I_μ = magnetizing component

(*) Transformer on No load :



① A transformer is on no-Load when it's secondary winding is open circuited i.e $I_2 = 0$

② under such a condition, the primary input current has to supply i) iron loss in the core & a very small amount of copper loss in the primary winding (the secondary winding being open involves no copper loss)

③ The primary winding in not pure winding and has some winding resistance.

④ Hence the no load primary input current (I_0) is not at $90°$ behind V_1 but lags it by an angle $\phi_0 < 90°$

⑤ Now the primary current (I_0) under no-Load condition can be split up into two components.

i) <u>iron loss</u> component is in phase with V_1

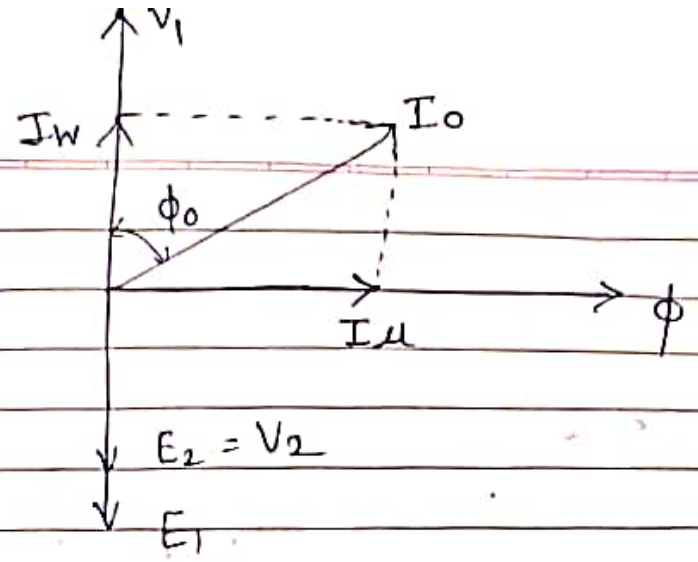
$$I_w = I_0 \cos\phi_0$$

ii) magnetizing component I_μ quadrature with V_1

$$I_\mu = I_0 \sin\phi_0$$

⑥ input power at no-Load condition is practically equal to iron loss in the transformer

Iron loss = $W_i = V_1 I_0 \cos\phi_0$



→ Since the flux is common to both the winding, ϕ is chosen as reference.

→ The self induced emf E_1 lags behind the flux ϕ by $90°$ hence the phasor flux E_1 lags behind the flux by $90°$.

→ the phasor E_1 is in phase with E_2.

→ the magnetizing current I_μ is drawn in phase with the flux ϕ.

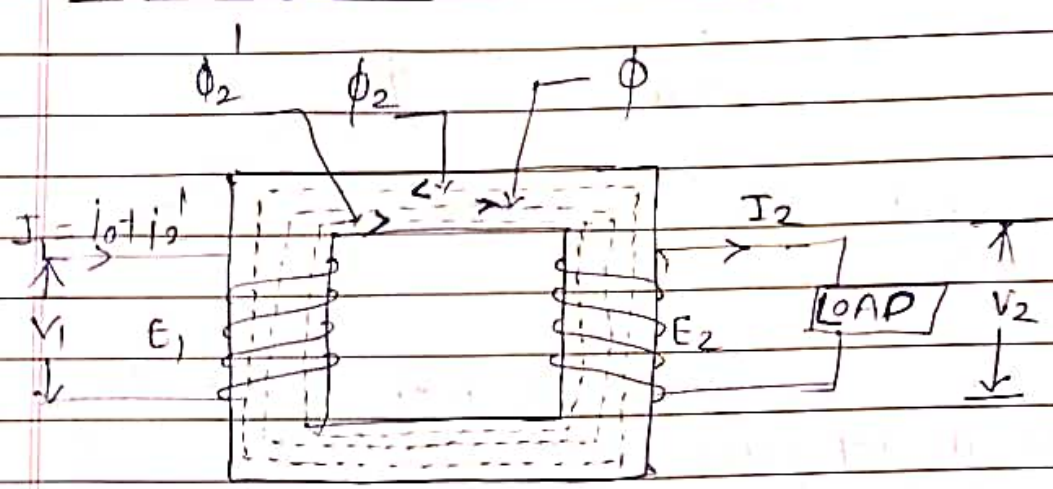
→ the applied voltage V_1 is drawn equal & opposite to E_1

→ the Active Component (Iron loss Component) is drawn in phase with V_1.

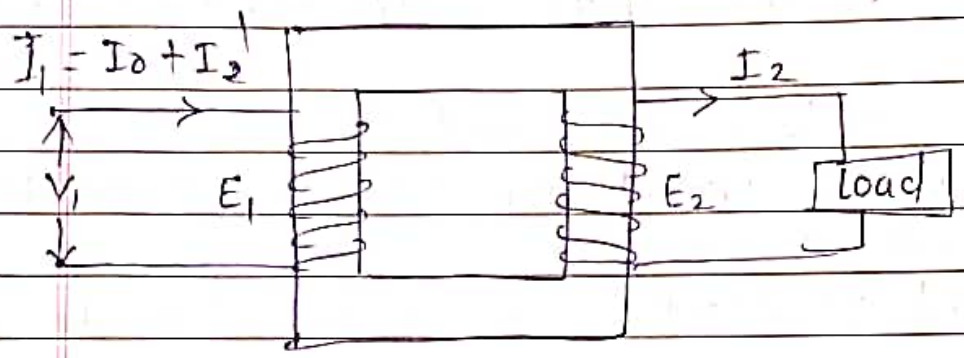
→ the phasor sum of I_W and I_μ gives the no load current I_0

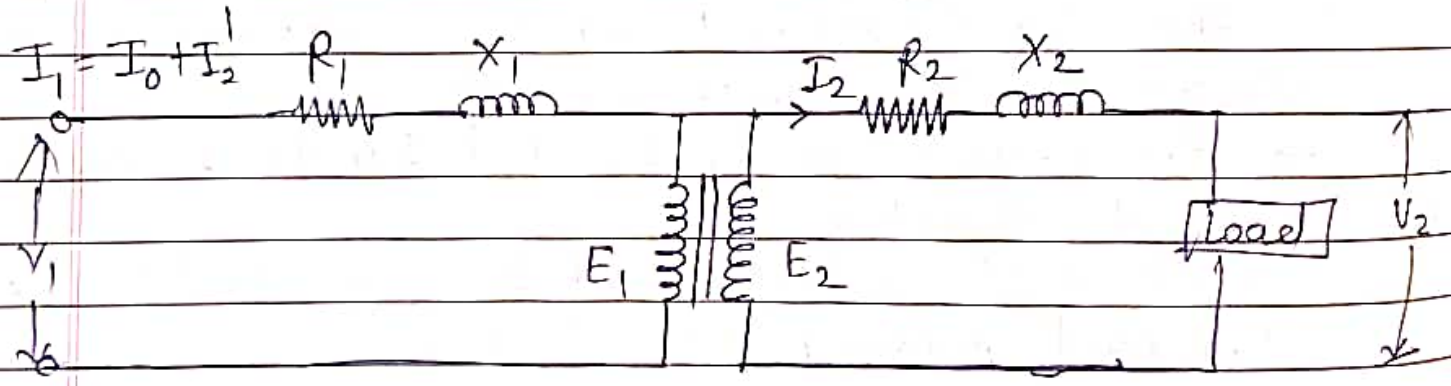
→ the angle between V_1 & I_0 is marked as ϕ_0 (no load power factor angle.)

(*) Transformer on load



$I_0 = I_w + I_\mu$





$$\overline{V_1} = \overline{I_1}\,\overline{R_1} + \overline{I_1}\,\overline{X_1} + (-\overline{E_1})$$

$$\& \; E_2 = \overline{I_2}\,\overline{R_2} + \overline{I_2}\,\overline{X_2} + \overline{V_2}$$

where

$$\overline{I_1} = \overline{I_0} + \overline{I_2'}$$

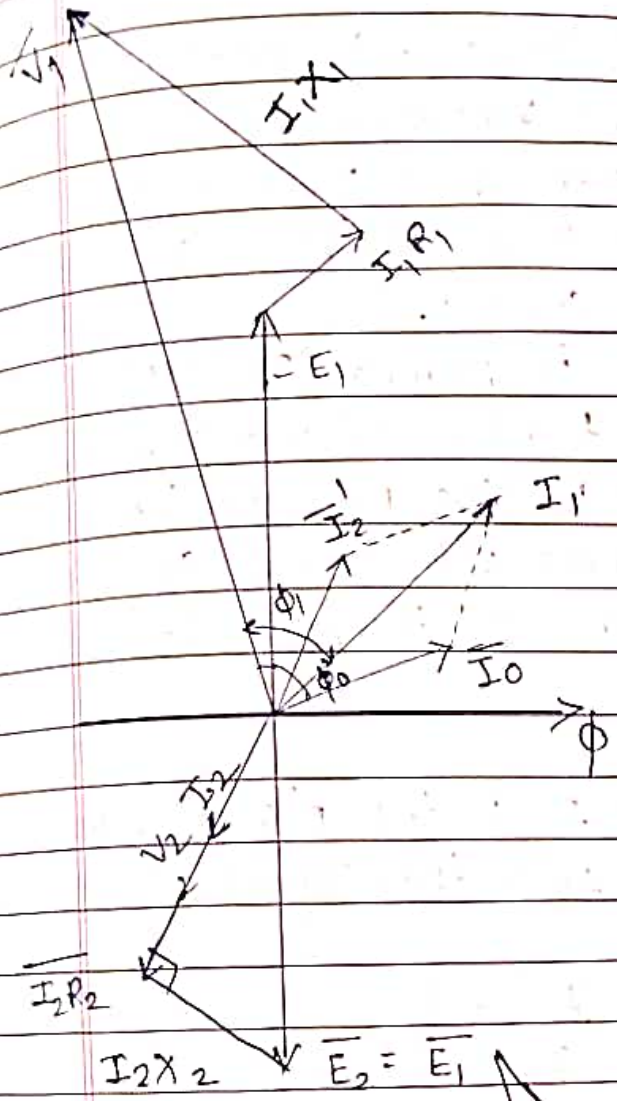


fig a) phasor diagram for Resistive

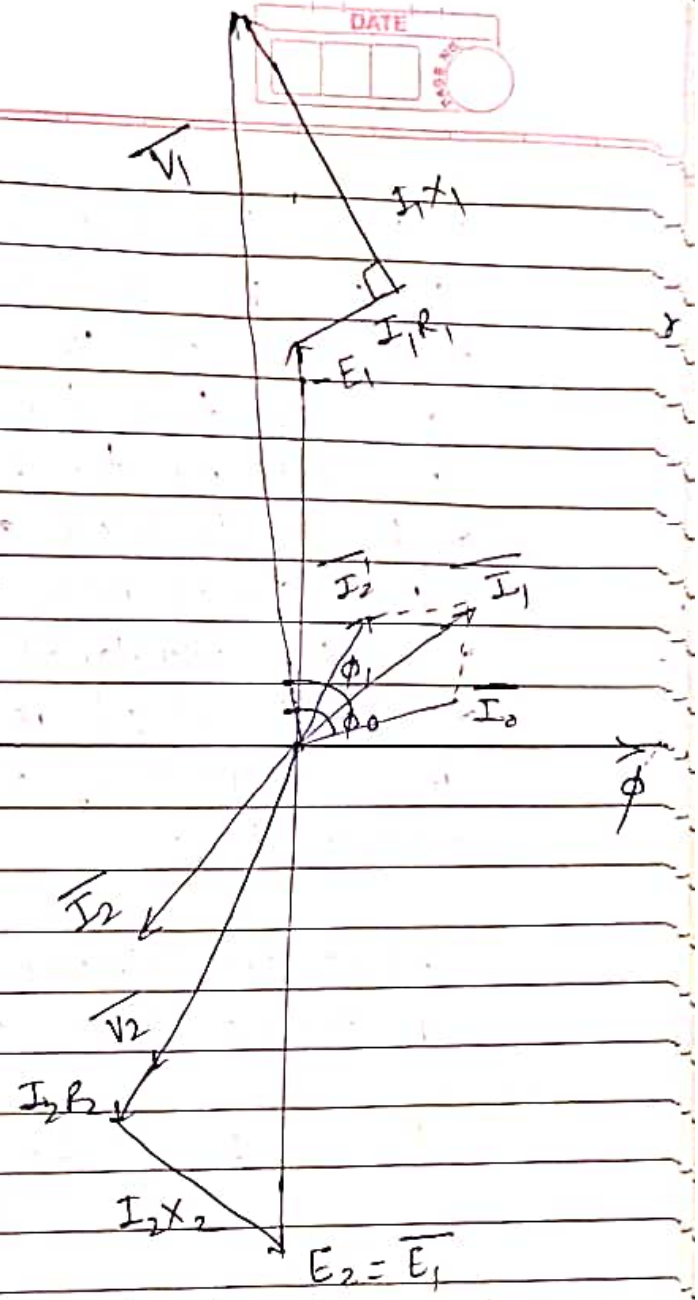


fig b) phasor diagram for inductive load

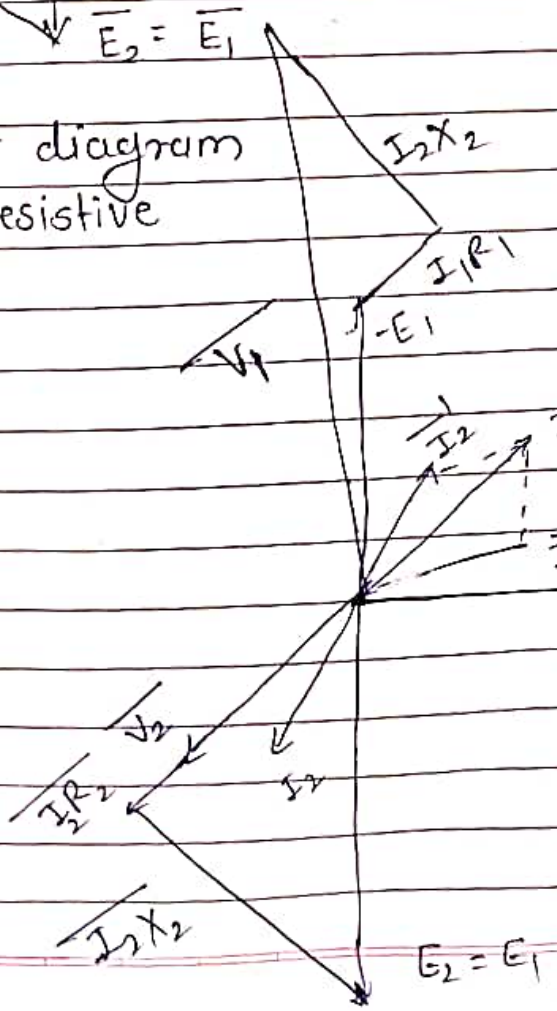


fig c) phasor diagram of Capacitive load

> when the transformer is loaded, a current I_2 will flow in the secondary winding.

> The secondary current I_2 set up a secondary flux ϕ_2 that tends to reduce the flux ϕ produced by the primary current.

> Hence induced emf E_1 in primary reduces. This causes more current to flow in the primary.

> let the additional current in the primary be I_2'.

> This current I_2' is anti phase with I_2 and set up it's own flux ϕ_2' which cancels the flux ϕ_2 produced by I_2

> Hence the primary current I_1 is the phasor sum of the no-load current I_0 & the current I_2'

$$\frac{N_2}{N_1} = \frac{I_1}{I_2} = \frac{I_0 + I_2'}{I_2} = \frac{I_2'}{I_2} = k$$

* Equivalent circuit of single phase Transformer

> The complete details of the current & their components, voltages, winding parameter of primary as well as secondary winding of a transformer are presented in one common electric circuit called equivalent circuit of Transformer.

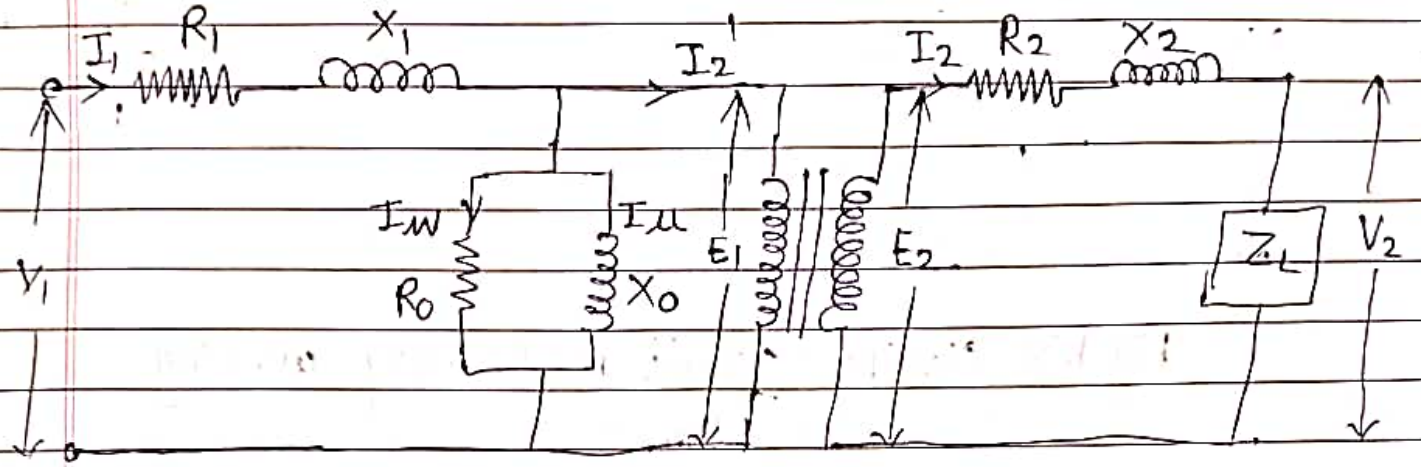


Fig.) Complete equivalent circuit of a practical Transformer.

> We know that no load current I_0 has two Components I_W & I_μ

> Therefor the equivalent circuit, I_0 is simulated by pure inductance X_0 taking the magnetizing Component I_μ

&

R_0 taking the working Component I_W connected in parallel across the primary.

> The Resistance R_0 represents the core loss and, so the current I_W which supplies core loss, is shown passing through it.

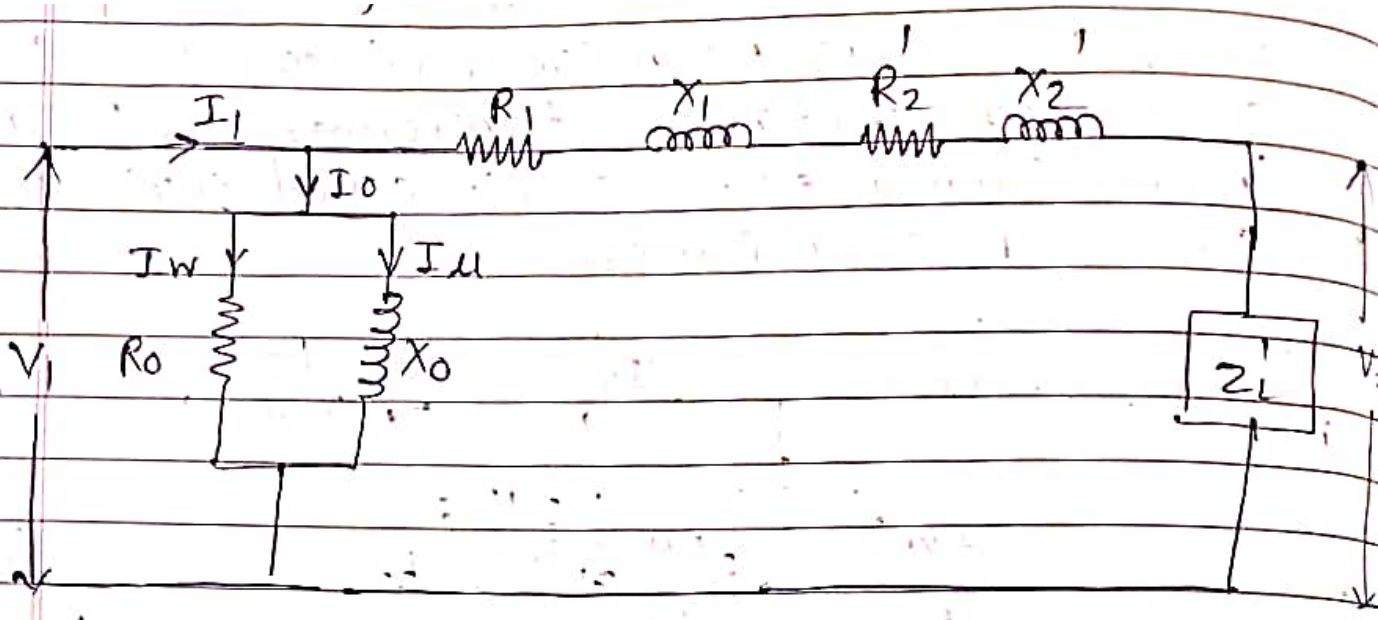
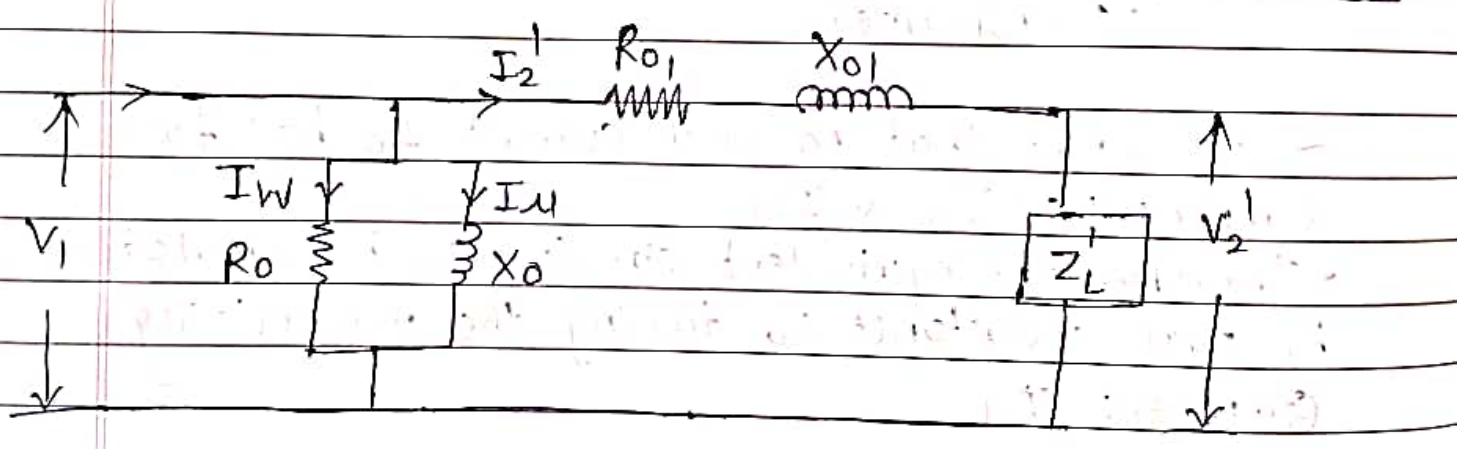


Fig b> Modify circuit for primary winding

⊛ equivalent circuit reffered to Primary winding



⇒ The equivalent resistance reffered to primary side

$$R_{01} = R_1 + R_2' = R_1 + \frac{R_2}{k^2}$$

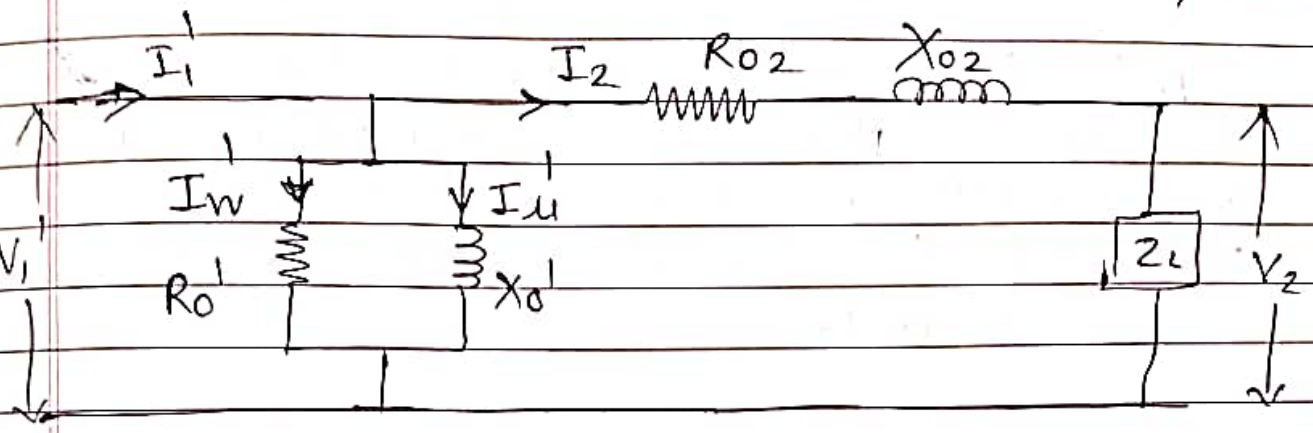
⇒ The equivalent reactance reffered to primary side

$$X_{01} = X_1 + X_2' = X_1 + \frac{X_2}{k^2}$$

⇒ The equivalent impedance reffered to primary side

$$Z_{01} = \sqrt{R_{01}^2 + X_{01}^2}$$

(*) equivalent circuit reffered to secondary side



- The equivalent resistance reffered to secondary side

$$R_{02} = R_2 + R_1' = R_2 + k^2 R_1 = k^2 R_{01}$$

- The equivalent reactance reffered to secondary side.

$$X_{02} = X_2 + X_1' = X_2 + k^2 X_1 = k^2 X_{01}$$

- The equivalent impedance reffered to secondary

$$Z_{02} = \sqrt{R_{02}^2 + X_{02}^2} = k^2 Z_{01}$$

Note i> while shifting any primary resistance or reactance to secondary, multiply it by k^2

ii> while shifting any secondary resistance or reactance to the primary, divide it by k^2

6.14 OPEN CIRCUIT (OC) TEST

[Dec 2014, 2015]

The purpose of this test is to determine (i) iron loss or core loss (W_i) (ii) magnetising resistance R_0, and (iii) magnetising reactance X_0.

Figure 6.21 shows the circuit diagram for conducting OC test on the transformer. In this test, one winding (usually high-voltage winding) is left open and the other winding is connected to a supply of normal voltage and frequency. An ammeter, voltmeter and wattmeter are connected on this side. The ammeter indicates no-load current drawn by the transformer. As the no-load current is usually 3 to 5% of the full-load current, copper losses are negligible and the wattmeter indicates iron loss.

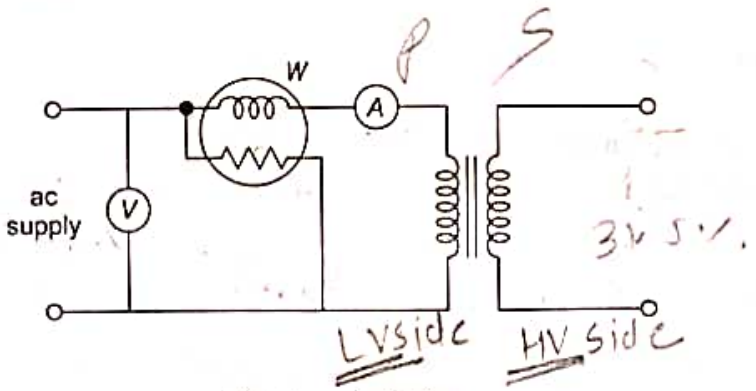


Fig. 6.21 O.C. test

Calculations

(i) When meters are connected on the primary side

Wattmeter reading = W_i

Voltmeter reading = V_1

Ammeter reading = I_0

$$W_i = V_1 I_0 \cos \phi_0$$

$$\cos \phi_0 = \frac{W_i}{V_1 I_0}$$

$$I_w = I_0 \cos \phi_0 \qquad I_\mu = I_0 \sin \phi_0$$

$$R_0 = \frac{V_1}{I_w}$$

$$X_0 = \frac{V_1}{I_\mu}$$

(ii) When meters are connected on the secondary side

Wattmeter reading = W_i

Voltmeter reading = V_2

Ammeter reading = I'_0

$$W_i = V_2 I'_0 \cos \phi'_0$$

$$\cos \phi'_0 = \frac{W_i}{V_2 I'_0}$$

$$I'_w = I'_0 \cos \phi'_0 \qquad I'_\mu = I'_0 \sin \phi'_0$$

$$R'_0 = \frac{V_2}{I'_w} \qquad X'_0 = \frac{V_2}{I'_\mu}$$

$$R_0 = \frac{R'_0}{K^2} \qquad X_0 = \frac{X'_0}{K^2}$$

6.15 SHORT-CIRCUIT (SC) TEST

[May 2015, Dec 2015]

The purpose of this test is to determine (i) full-load copper loss, (ii) equivalent resistance R_{01} or R_{02}, and (iii) equivalent reactance X_{01} or X_{02}.

Figure. 6.22 shows the circuit diagram for conducting an SC test on the transformer. In this test, one winding (usually low-voltage winding) is short circuited, while a low voltage is applied to the other winding. The applied voltage is slowly increased until full-load current flows in this winding and hence, through the other winding. Normally, the applied voltage is 5 to 10% of the rated voltage of this winding. Hence, fluxes produced in the core are small and the iron losses are very small. Thus, wattmeter indicates full-load copper loss.

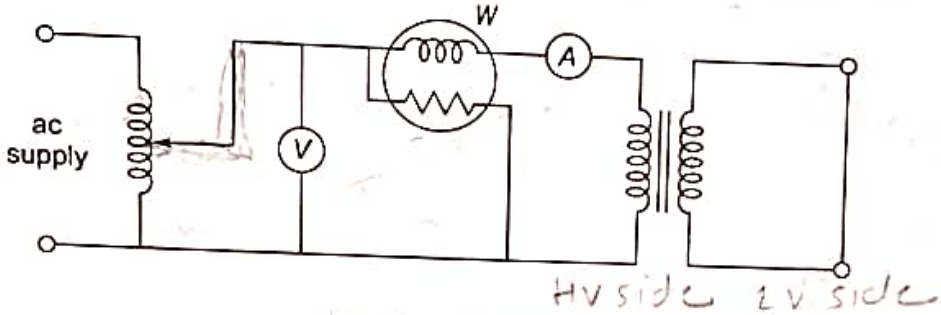


Fig. 6.22 SC test

Calculations

Wattmeter reading $= W_{sc}$

Voltmeter reading $= V_{sc}$

Ammeter reading $= I_{sc}$

(i) When meters are connected on the primary side

$$Z_{01} = \frac{V_{sc}}{I_{sc}}$$

$$R_{01} = \frac{W_{sc}}{I_{sc}^2}$$

$$X_{01} = \sqrt{(Z_{01})^2 - (R_{01})^2}$$

(ii) When meters are connected on the secondary side

$$Z_{02} = \frac{V_{sc}}{I_{sc}}$$

$$R_{02} = \frac{W_{sc}}{I_{sc}^2}$$

$$X_{02} = \sqrt{(Z_{02})^2 - (R_{02})^2}$$

Efficiency of a transformer -

The efficiency of any device is defined as the ratio of the output power to the input power. The efficiency of a transformer is expressed as,

$$\eta = \frac{\text{Output power}}{\text{Input power}} = \frac{\text{output power}}{\text{output power} + \text{Total losses}}$$

$$= \frac{\text{output power}}{\text{output power} + W_i + W_{cu}}$$

where, W_i = iron loss & W_{cu} = copper loss

- The efficiency of a transformer at any load is given by;

$$\% \eta = \frac{(x \times \text{full load kVA} \times PF)}{(x \times \text{full load kVA} \times PF) + W_i + x^2 [W_{cu}]_{FL}}$$